Pre-work · Axial Skeleton

Axial Skeleton, the skull and the spine. Read both Part A chapters first. This is the largest volume of naming in the course, so the organizing is what makes it survivable.

Module 2 · 2 of 4 | Bring this to class

How to work this pre-work

The order matters more than the hours. Context first, content second. Most students who struggle did every piece of this, in the wrong sequence.

1. **Read**
   Read the Part A chapter first. This packet does not repeat it. It puts it to work.
2. **Organize**
   Panel 1. Turn the reading into mini visuals. Not copied sentences, your structure.
3. **Draw**
   Panel 2. Brain dump from memory, then draw, then correct in a second colour.
4. **Apply**
   Panel 3. Notes closed for the first pass. Open them only to check.
5. **Retrieve**
   Later in the week, not the same night. Cover everything and redraw your mini visuals from memory. What you cannot redraw is what you do not yet know.

Why this order. Reading before you organize gives you context. Organizing before you draw forces you to decide what connects to what. Drawing before you apply is where you find out what you actually know. If you skip to the application questions, you will answer them with the notes open and learn almost nothing.

PANEL 1 OF 3

Organize it

Turn the skull and spine chapters into mini visuals. Eight chunks. The grids do the heavy lifting here.

FIVE SHAPES, THAT IS THE WHOLE TOOLKIT

Information has a shape. Pick the wrong one and the visual is noise. Pick the right one and you can redraw it from memory in twenty seconds. Ladder for an ordered series. Chain for a sequence, with arrows. Split for a pair you keep confusing. Hub for one thing with parts hanging off it. Grid for two variables crossed.

Your chunks for this packet

Each chunk gets one visual, in the shape that fits it, drawn on the mini visual sheet at the end of this packet. Each is made once, so nothing here is done twice. If a chunk

PANEL 2 OF 3

Draw it

Panel 1 was the shape of the information. This is the structure itself, drawn from memory, then corrected.

HOW A BRAIN DUMP WORKS

Set a timer for two minutes. Close everything, write and sketch every piece you can recall on blank paper. Do not organize it. When the timer stops, then draw. The gap between the dump and the drawing is the part that teaches you.

PANEL 3 OF 3

Apply it

First pass with everything closed. Then open the notes in a second colour and mark what you missed.

WORKED EXAMPLE

A loose vertebra has a transverse foramen on each transverse process. Name the region and say how you know.

Answer. Cervical.

Reasoning. Go to the giveaway feature first, not the size or the shape. Only cervical vertebrae have transverse foramina, because only they carry the vertebral arteries up to the brain. Thoracic vertebrae are placed by costal facets, and lumbar by their heavy body and blunt spinous process. So one feature settles it, and the other features are only confirmation.

Set A · Name the bone and the opening

1. The cribriform plate belongs to the ______________________ and passes ______________________.

takes longer than three minutes you are copying instead of organizing.

1 **SPLIT**

Cranial against facial bones

One split, eight on one branch and fourteen on the other. Mark which are single and which are paired. This is the frame for the whole skull.

2 **GRID**

The four sutures

Suture down the side, bones joined and location across the top. Add the pterion as a fifth row and mark why it matters.

3 **GRID**

The major foramina

Foramen down the side, bone and what passes through across the top. Nine rows. Star the four you will confuse.

4 **HUB**

The sphenoid

Sphenoid at the hub, because it touches every other cranial bone. Spoke to body, sella turcica, greater and lesser wings, pterygoid processes, optic canal, superior orbital fissure.

5 **GRID**

The four fontanelles

Fontanelle down the side, location and closing age across the top. Four rows, and the ages are the part people lose.

6 **LADDER**

The five regions of the spine

Five rungs, cervical at the top. Write the count beside each rung and the two fused regions at the bottom.

DRAWING 1

The skull, lateral view

BRAIN DUMP FIRST

Two minutes. Every bone you can name on the side of the skull, and the sutures between them.

Draw the skull in profile large enough to label. Outline each bone visible from the side and shade them differently: frontal, parietal, temporal, occipital, sphenoid, zygomatic, maxilla, nasal, lacrimal, mandible. Draw the four major sutures. Mark the pterion where four bones meet, the mastoid and styloid processes, the external acoustic meatus, and the zygomatic arch.

In your notebook, head the page **M2–1 The skull, lateral view** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can put your finger on the pterion and say which four bones meet there and what runs beneath it.

2. The pituitary gland sits in the ____________________ of the ____________________.
3. The foramen magnum passes the ____________________ and the ____________________.
4. The only freely movable bone of the skull is the ____________________.
5. Name the seven bones that build the orbit.

____

____

Set B · Predict it

1. A blow lands on the pterion. Predict the injury and name the vessel involved.

____

____

2. An infant has a bulging anterior fontanelle. Say what it may signal, and what a sunken one may signal instead.

____

____

3. A tooth infection in the upper jaw spreads. Name the cavity most at risk and say why.

____

____

4. A patient has a herniated disc at L4 to L5 with pain travelling down the leg. Name the two parts of the disc and explain the path of the pain.

____

____

Set C · Sort and classify

1. Sort as true, false, or floating: ribs 3, 9, 11.

____

2. Name the four curvatures and mark each primary or secondary, with the milestone that produces the secondary ones.

____

____

7 **GRID**

Cervical, thoracic, lumbar vertebrae

Region down the side. Body shape, spinous process, and the one giveaway feature across the top. Three rows. The giveaway column is what lets you place a loose vertebra.

8 **CHAIN**

The four curvatures, in developmental order

Thoracic and sacral present at birth, then cervical when the head lifts, then lumbar when the child walks. Arrows carry the milestones.

YOU KNOW PANEL 1 WORKED WHEN

Someone hands you a blank page and you can redraw any one of these from memory in under a minute, explaining it out loud while you draw.

DRAWING 2

A typical vertebra, from above

BRAIN DUMP FIRST

One minute. Every part of a vertebra, and which region it belongs to.

Draw one vertebra looking down on it. Body at the front, then the pedicles, laminae, and the arch closing behind, with the vertebral foramen in the middle. Add the spinous process pointing back and the transverse processes to the sides, then the superior articular processes. Now redraw it twice more in a second colour, once as cervical with transverse foramina and once as thoracic with costal facets.

In your notebook, head the page **M2–2 A typical vertebra, from above** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

Someone hands you a loose vertebra and you can name its region from one feature without hesitating.

3. Compare the atlanto-occipital and atlanto-axial joints by the movement each produces.

YOU KNOW PANEL 3 WORKED WHEN

Your first-pass answers and your corrections are in two different colours, and you can say out loud why each correction was a correction.

DRAWING 3

The thoracic cage

BRAIN DUMP FIRST

One minute. The ribs, how they attach, and the parts of the sternum.

Draw the cage from the front. Sternum in three parts with the jugular notch, sternal angle, and xiphoid process labelled. Draw the twelve rib pairs and their costal cartilages, then bracket them: true, false, and floating. Mark the sternal angle at the level of the second rib.

In your notebook, head the page **M2–3 The thoracic cage** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can count ribs from the sternal angle the way a clinician does, and say where you would stop for each group.

Mini visual sheet

One frame per chunk, shaped for what goes in it. Tall frames are ladders, wide strips are chains, squares are hubs. Draw straight onto this sheet. If a frame is too small for your handwriting, redraw that one in your notes and put the page number on the line.

1 Cranial against facial bones

SPLIT

2 The four sutures

GRID

3 The major foramina

GRID

4 The sphenoid

HUB

5 The four fontanelles

GRID

6 The five regions of the spine

LADDER

7 Cervical, thoracic, lumbar vertebrae

GRID

8 The four curvatures, in developmental order

CHAIN